

Working with Real Serial Ports using GTKTerm

Requisitos de conclusão

This page explains how to configure and use real serial ports in Linux, in the context of the Lab 1 protocol work.

GTKTerm is a Linux utility that allows **sending and receiving bytes through the serial port using an interactive shell**.

It is recommended using GTKTerm when working with real serial ports in the laboratory.

Configure the Serial Port Parameters in GTKTerm

To use GTKTerm, you must first configure the parameters of the serial port, namely:

- **Serial port device:** The serial port device name mounted in Linux (e.g., /dev/ttyS0).
- **Baudrate:** The number of symbols sent / received per second (e.g, 9600 bps).

NOTE: Both the transmitter and the receiver must have exactly the same serial port parameters. Otherwise, the bytes will not be received correctly.

Configure the Representation of Bytes in GTKTerm

By default, GTKTerm shows the bytes in ASCII format. To view the received bytes in hexadecimal format, select View > Hexadecimal in the GTKTerm toolbar.

Note that GTKTerm only shows the hexadecimal characters. It does not show the 0x prefix. For example, the FLAG byte (0x7E) is represented in GTKTerm as 7E.

Sending / Receiving Bytes via the Serial Port using GTKTerm

GTKTerm is useful when working with real serial ports, since it allows sending bytes interactively with the keyboard and visualising them in the receiver. GTKTerm can be used in two modes:

- In the transmitter PC to send bytes interactively using the keyboard.
- In the receiver PC to show the bytes received in the serial port.

Specifically, GTKTerm can be used to check if the serial port is working correctly. It is recommended to check that the serial port is working correctly before starting to work on the protocol. This ensures that all problems are due to your code and not the serial port hardware.

To check if the serial port is working correctly, follow these steps:

- Open GTKTerm in the transmitter PC and the receiver PC.
- Configure both GTKTerm programs with the same serial port parameters.
- In the transmitter PC, write letters in GTKTerm to send them to the serial port.
- In the receiver, watch those letters appear in the terminal.
- If the bytes sent by the transmitter in GTKTerm are not received in the receiver, it means the serial port is not working correctly. This may be due to a mismatch of the serial port parameters in the transmitter and receiver PCs or an error in the serial port itself.